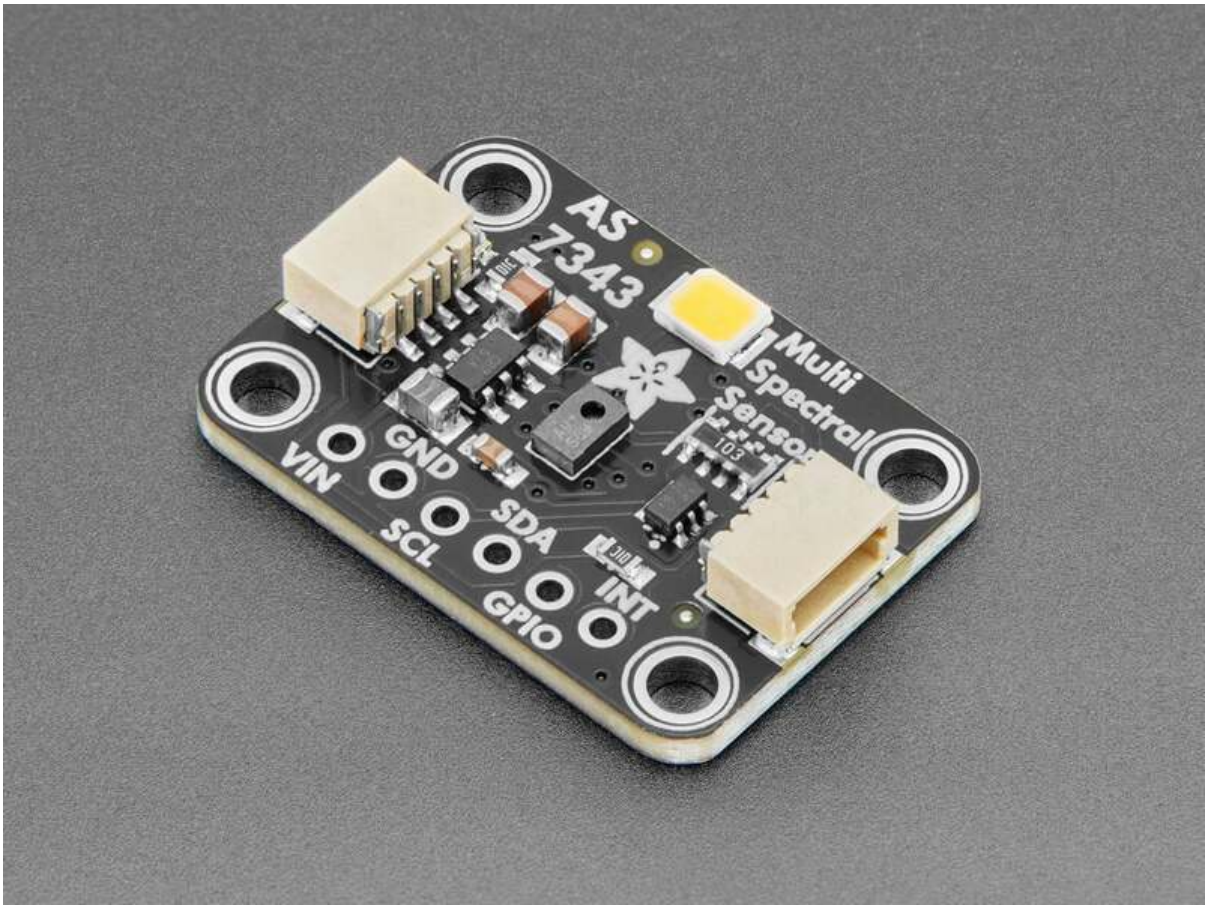
Adafruit AS7343 14-Channel Light / Color Sensor Breakout

Created by Tim C



https://learn.adafruit.com/adafruit-as7343-14-channel-light-color-sensor-breakout

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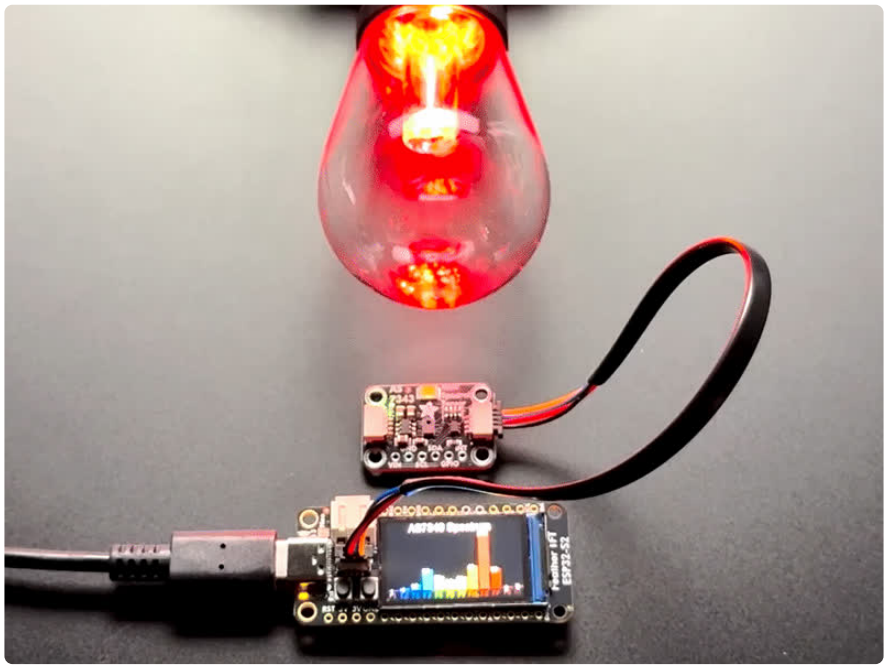
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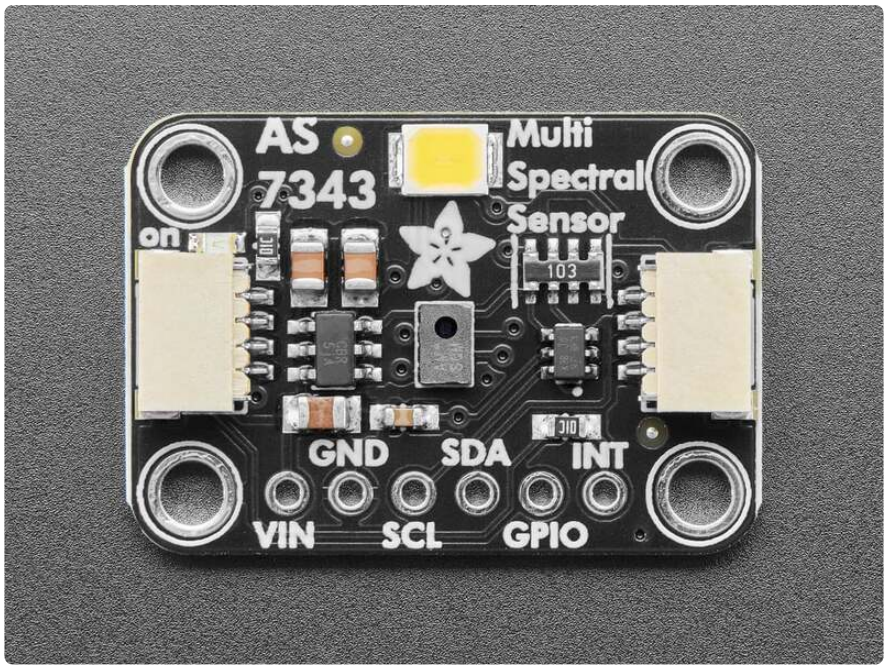
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Overview

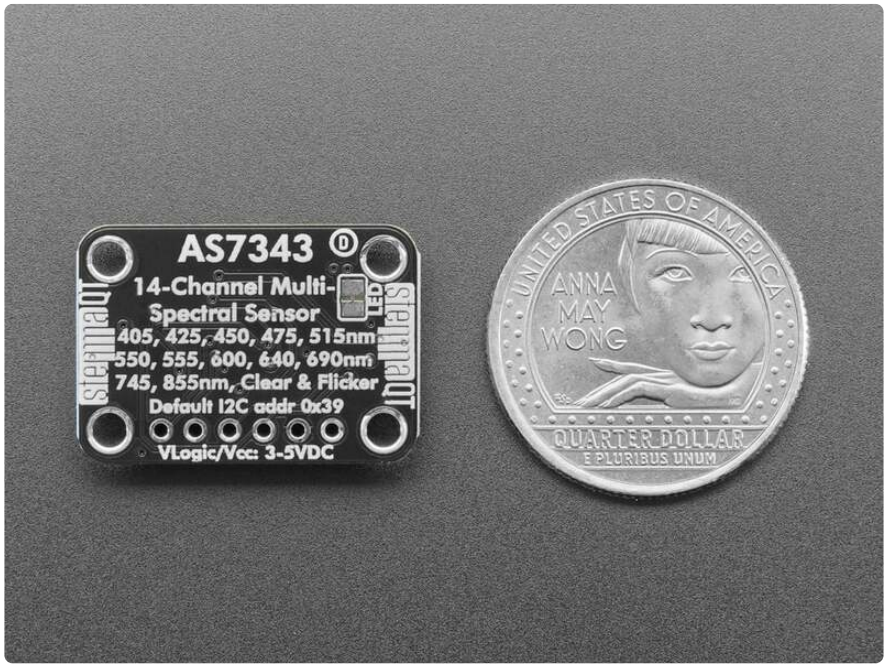


The **Adafruit AS7343 14-Channel Light / Color Sensor Breakout** is a multi-channel spectrometer, which is a special type of light sensor that is able to detect not only the amount of light present but also the amounts of light within different wavelengths. You can use it determining what color or colors of light is present much better than the human eye is capable of.

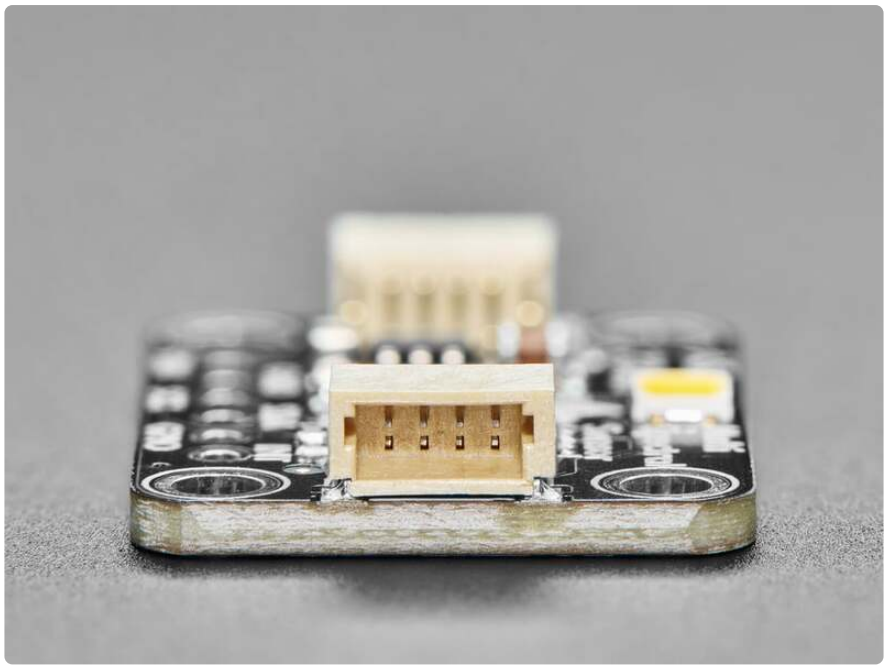


The AS7343 packs within its 3x2mm footprint 25 photodiodes that can detect 11 separate, overlapping bands of visible light, plus one near infrared, one clear and a flicker channel for 14 channels total. Compared to the AS7341, this sensor has 4 more light channels - but you'll need to use a new firmware driver since it's not backward-compatible.

The multi-spectral color measuring capabilities of the AS7343 can be used to quantify the specific makeup of light, either emitted or reflective. It's kind of like a mini spectroscopy sensor. This is possible thanks to the impressive collection of sensors in the AS7343 being routed through a 16-bit 6-channel ADC that takes the raw measurements and converts them into digital values that can be read out over I2C.

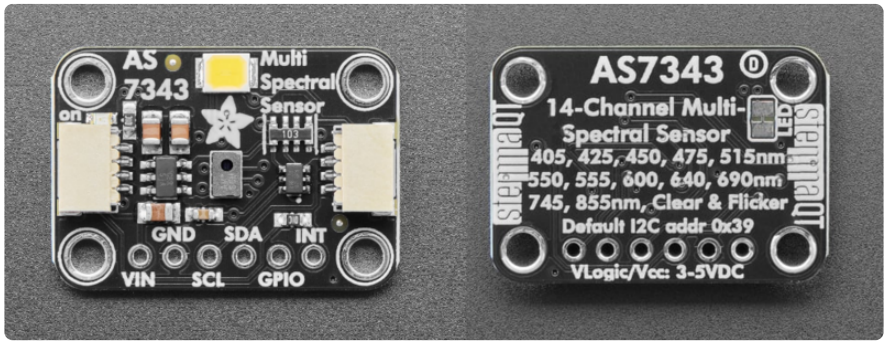


14 readable individual sensor elements (13 light channels plus flicker detection) don’t exactly fit through a 6-channel ADC all at once, so the chip includes a so-called Super MUX (SMUX) that allows you to route the signal from any sensor to any ADC channel. The sensor also has GPIO and interrupt pins that can allow it to communicate directly with other sensors, or the microcontroller itself.



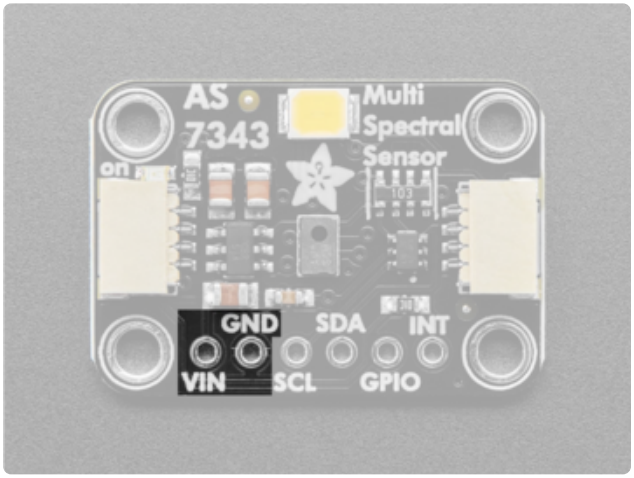
All of this capability is made accessible by mounting the sensor on a STEMMA QT form factor breakout board, complete with level shifting circuitry and SparkFun Qwiic (https://adafru.it/Fpw) compatible Stemma QT (https://adafru.it/Ft4) connectors. This means that you can, without needing to solder, connect our AS7343 breakout into your 3.3V or 5V microcontroller of choice be it an Arduino Uno, Raspberry Pi, or one of the many CircuitPython-compatible boards (https://adafru.it/Em8). While it certainly takes a bit of work to make all those different light sensors share their measurements, our Arduino and CircuitPython libraries take care of all of that hard work for you and even include example code to help get you started. **QT Cable is not included**, but we have a variety in the shop (https://adafru.it/17VE).

Pinouts



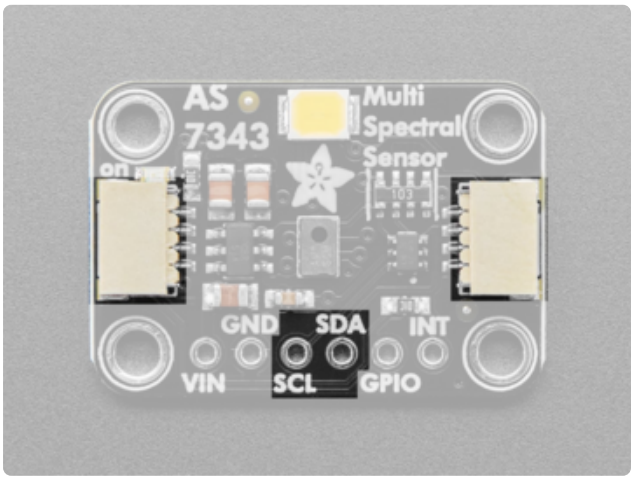
The fixed I2C address is **0x39**.

Power Pins



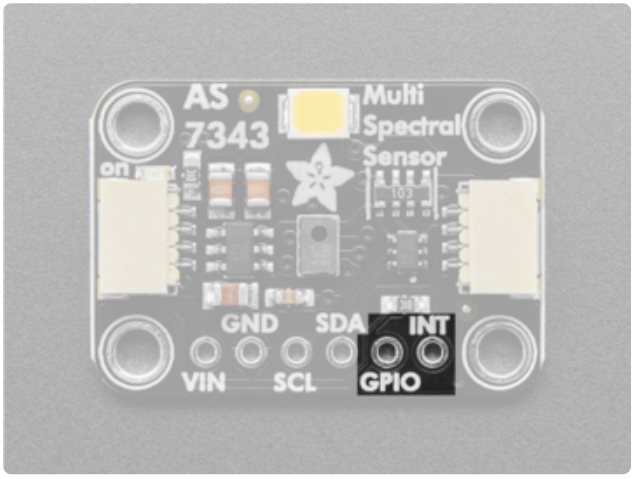
VIN - this is the power pin. Since the sensor chip uses 3VDC, we have included a voltage regulator on board that will take 3-5VDC and safely convert it to 3VDC. To power the board, give it the same power as the logic level of your microcontroller - e.g. for a 5V micro like Arduino, use 5V.
GND - common ground for power and logic

I2C Data Pins

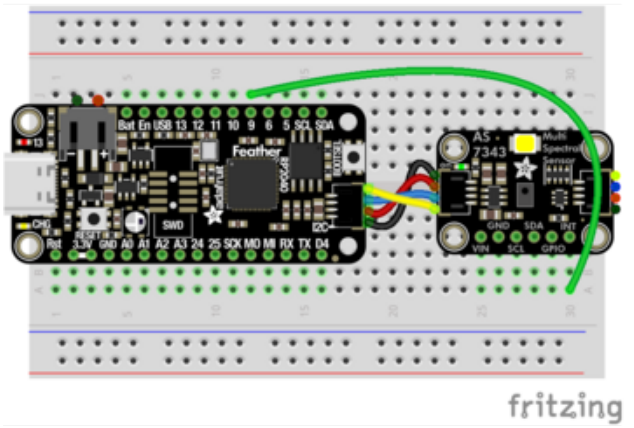


SCL - the I2C clock pin, connect to your microcontroller's I2C clock line.
SDA - the I2C data pin, connect to your microcontroller's I2C data line.
STEMMA QT **(https://adafru.it/Ft4)** - These connectors allow you to connectors to dev boards with **STEMMA QT** connectors or to other things with various associated accessories (https://adafru.it/Ft6)

Interrupt & GPIO Pins

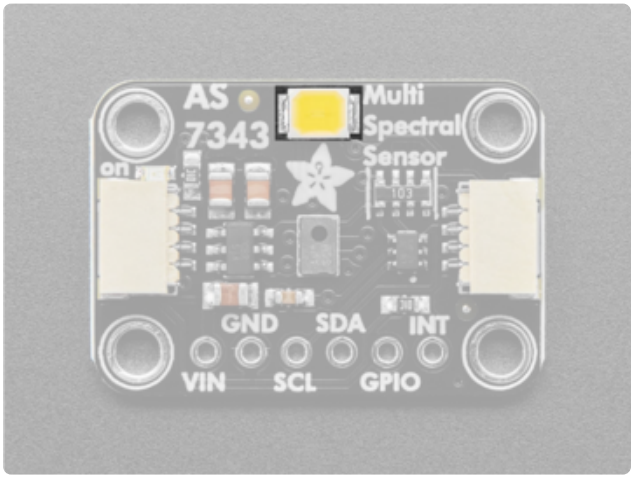


INT - the interrupt pin. It is an open-drain interrupt. To use, connect the pin to a GPIO pin on your microcontroller.



GPIO - the GPIO pin on the breakout. It is an open-drain pin that can be used as an input or output.

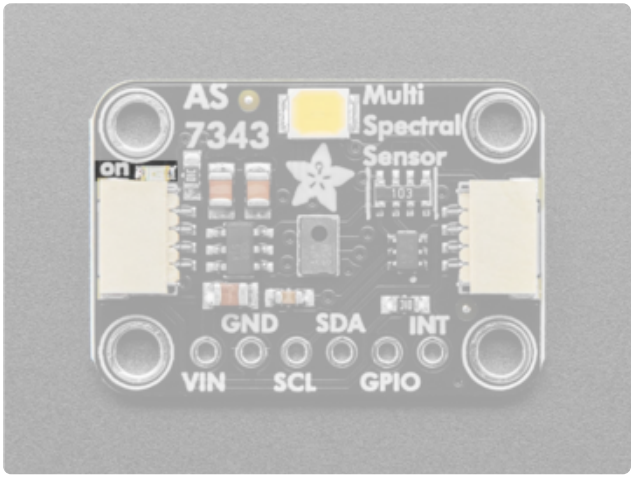
White LED

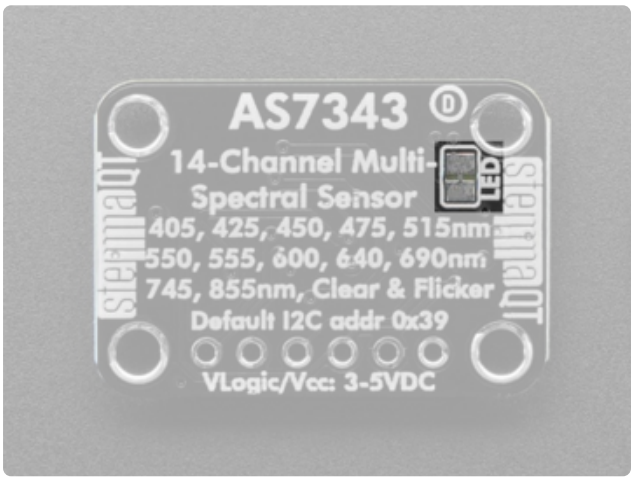


The AS7343 includes LED driver circuitry. This breakout has a white LED connected to the driver. It can be controlled using the CircuitPython or Arduino libraries.

This can be used to shine white light onto an object and measure the colors in the light being reflected back from the object into the sensor.

Power LED & Jumper





Above the left STEMMA QT connector there is a small green **ON** power indicator LED. When the breakout is powered this LED will turn on.

On the back of the AS7343 breakout there is a jumper with a small trace between two pads that can be cut to disable the **ON** power indicator LED on the board.

If you do not want that indicator lit, carefully cut the trace. It can always be reconnected with a bit of solder.

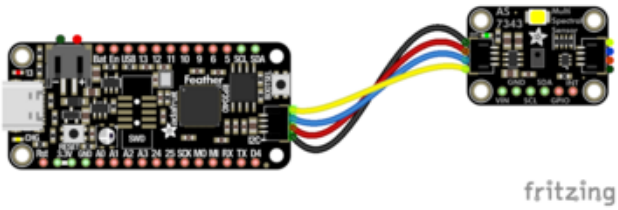
CircuitPython and Python

It's easy to use the **AS7343** with Python or CircuitPython, and the Adafruit_CircuitPython_AS7343 (https://adafru.it/1aB8) module. This module allows you to easily write Python code to read data from the sensor.

You can use this driver with any CircuitPython microcontroller board or with a computer that has GPIO and Python thanks to Adafruit_Blinka, our CircuitPython-for-Python compatibility library (https://adafru.it/BSN).

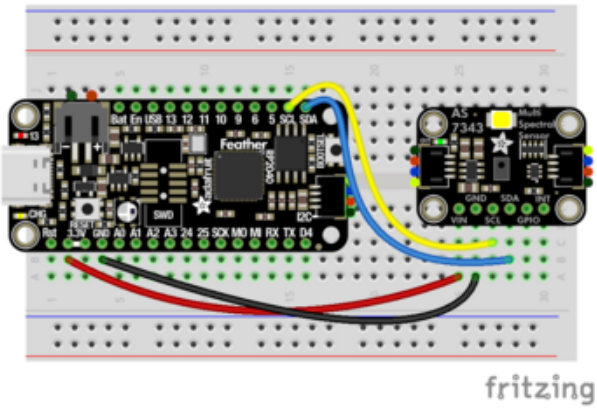
CircuitPython Microcontroller Wiring

First wire up a AS7343 to your board exactly as shown below. Here's an example of wiring a Feather RP2040 to the sensor with I2C using one of the handy **STEMMA QT** (https://adafru.it/Ft4) connectors:



Board STEMMA 3V to **sensor VIN (red wire)**
Board STEMMA GND to **sensor GND (black wire)**
Board STEMMA SCL to **sensor SCL (yellow wire)**
Board STEMMA SDA to **sensor SDA (blue wire)**

The following is the AS7343 wired to a Feather RP2040 using a solderless breadboard:

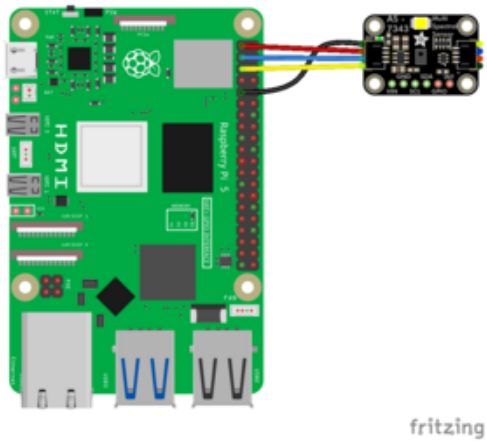


Board 3V to **sensor VIN (red wire)**
Board GND to **sensor GND (black wire)**
Board SCL to **sensor SCL (yellow wire)**
Board SDA to **sensor SDA (blue wire)**

Python Computer Wiring

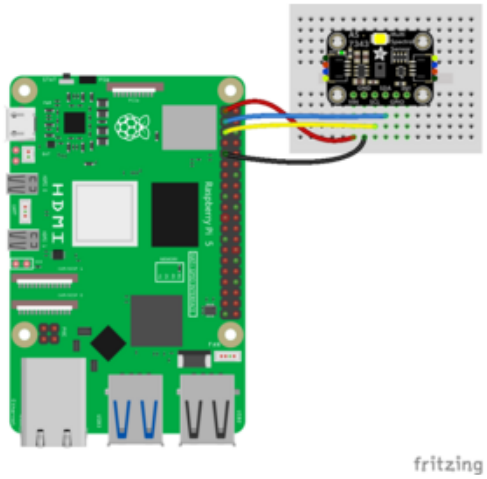
Since there are dozens of Linux computers/boards you can use, we will show wiring for Raspberry Pi. For other platforms, [please visit the guide for CircuitPython on Linux to see whether your platform is supported](https://adafru.it/BSN) (https://adafru.it/BSN).

Here's the Raspberry Pi wired with I2C using the STEMMA connector:



Pi 3V to **sensor VIN (red wire)**
Pi GND to **sensor GND (black wire)**
Pi SCL to **sensor SCL (yellow wire)**
Pi SDA to **sensor SDA (blue wire)**

Here's the Raspberry Pi wired with I2C using a solderless breadboard:



Pi 3V to **sensor VIN (red wire)**
Pi GND to **sensor GND (black wire)**
Pi SCL to **sensor SCL (yellow wire)**
Pi SDA to **sensor SDA (blue wire)**

Python Installation of AS7343 Library

You'll need to install the **Adafruit_Blinka** library that provides the CircuitPython support in Python. This may also require enabling I2C on your platform and verifying you are running Python 3. Since each platform is a little different, and Linux changes often, please visit the CircuitPython on Linux guide to get your computer ready (https://adafru.it/BSN)!

Once that's done, from your command line run the following command:

- `pip3 install adafruit-circuitpython-as7343`

If your default Python is version 3 you may need to run 'pip' instead. Just make sure you aren't trying to use CircuitPython on Python 2.x, it isn't supported!

CircuitPython Usage

To use with CircuitPython, you need to first install the **Adafruit_CircuitPython_AS7343** library, and its dependencies, into the **lib** folder on your **CIRCUITPY** drive. Then you need to update **code.py** with the example script.

Thankfully, we can do this in one go. In the example below, click the **Download Project Bundle** button below to download the necessary libraries and the **code.py** file in a zip file. Extract the contents of the zip file, and copy the **entire lib folder** and the **code.py** file to your **CIRCUITPY** drive.

Your **CIRCUITPY/lib** folder should contain the following folders and files:

- **adafruit_bus_device/**
- **adafruit_register/**
- **adafruit_as7343.mpy**



Python Usage

Once you have the library `pip3` installed on your computer, copy or download the following example to your computer, and run the following, replacing **code.py** with whatever you named the file:

```
python3 code.py
```

Example Code

If running CircuitPython: Once everything is saved to the **CIRCUITPY** drive, [connect to the serial console](https://adafru.it/Bec) (https://adafru.it/Bec) to see the data printed out!

If running Python: The console output will appear wherever you are running Python.

```
# SPDX-FileCopyrightText: Copyright (c) 2026 Tim Cocks for Adafruit Industries
#
# SPDX-License-Identifier: MIT
import time

import board

from adafruit_as7343 import AS7343

i2c = board.I2C()
sensor = AS7343(i2c)

CHANNEL_LABELS = [
    "FZ (450nm blue)",
    "FY (555nm yellow-green)",
    "FXL (600nm orange)",
    "NIR (855nm near-IR)",
    "VIS_TL_0 (clear top-left, cycle 1)",
    "VIS_BR_0 (clear btm-right, cycle 1)",
    "F2 (425nm violet-blue)",
    "F3 (475nm blue-cyan)",
    "F4 (515nm green)",
    "F6 (640nm red)",
    "VIS_TL_1 (clear top-left, cycle 2)",
    "VIS_BR_1 (clear btm-right, cycle 2)",
    "F1 (405nm violet)",
    "F7 (690nm deep red)",
    "F8 (745nm near-IR edge)",
    "F5 (550nm green-yellow)",
    "VIS_TL_2 (clear top-left, cycle 3)",
    "VIS_BR_2 (clear btm-right, cycle 3)",
]

while True:
    readings = sensor.all_channels
    print("--- AS7343 Channel Readings ---")
    for label, value in zip(CHANNEL_LABELS, readings):
        print(f"  {label}: {value}")
    print()
    time.sleep(1)
```

First, the sensor gets initialized over I2C. In the main loop the spectral data is read from the sensor and the different channel values are printed to the serial console once per second.

```
code.py output:
--- AS7343 Channel Readings ---
  FZ (450nm blue): 419
  FY (555nm yellow-green): 707
  FXL (600nm orange): 656
  NIR (855nm near-IR): 555
  VIS_TL_0 (clear top-left, cycle 1): 1151
  VIS_BR_0 (clear btm-right, cycle 1): 17999
  F2 (425nm violet-blue): 203
  F3 (475nm blue-cyan): 389
  F4 (515nm green): 493
  F6 (640nm red): 612
  VIS_TL_1 (clear top-left, cycle 2): 1151
  VIS_BR_1 (clear btm-right, cycle 2): 17999
  F1 (405nm violet): 90
  F7 (690nm deep red): 556
  F8 (745nm near-IR edge): 238
  F5 (550nm green-yellow): 196
  VIS_TL_2 (clear top-left, cycle 3): 1151
  VIS_BR_2 (clear btm-right, cycle 3): 17999
```

Python Docs

Python Docs (https://adafru.it/1aAV)

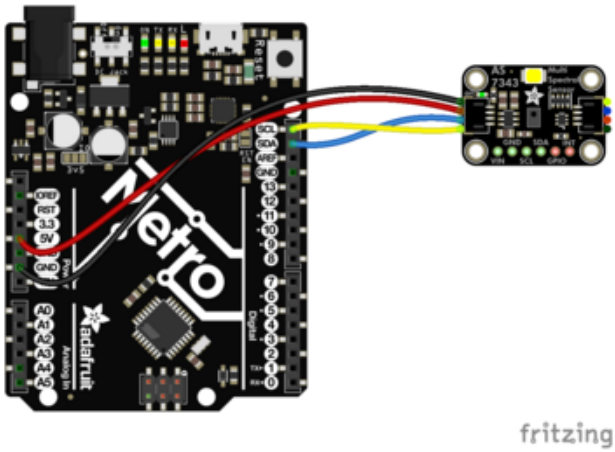
Arduino

Using the AS7343 breakout with Arduino involves wiring up the breakout to your Arduino-compatible microcontroller, installing the Adafruit_AS7343 (https://adafru.it/1aB9) library, and running the provided example code.

Wiring

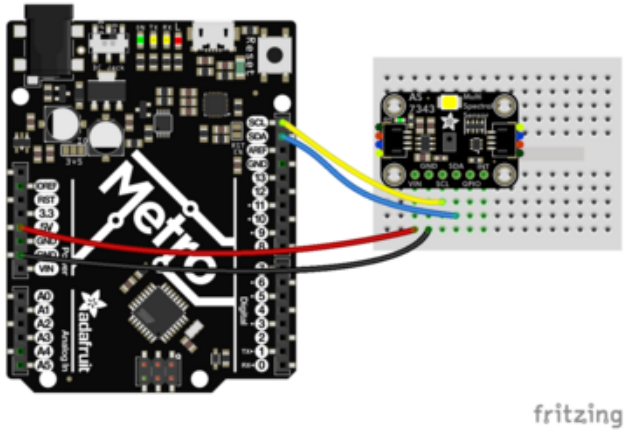
Wire as shown for a **5V** board like an Uno. If you are using a **3V** board, like an Adafruit Feather, wire the board's 3V pin to the sensor **VIN**.

Here is an Adafruit Metro wired up to the sensor using the STEMMA QT connector:



Board 5V to **breakout VIN (red wire)**
Board GND to **breakout GND (black wire)**
Board SCL to **breakout SCL (yellow wire)**
Board SDA to **breakout SDA (blue wire)**

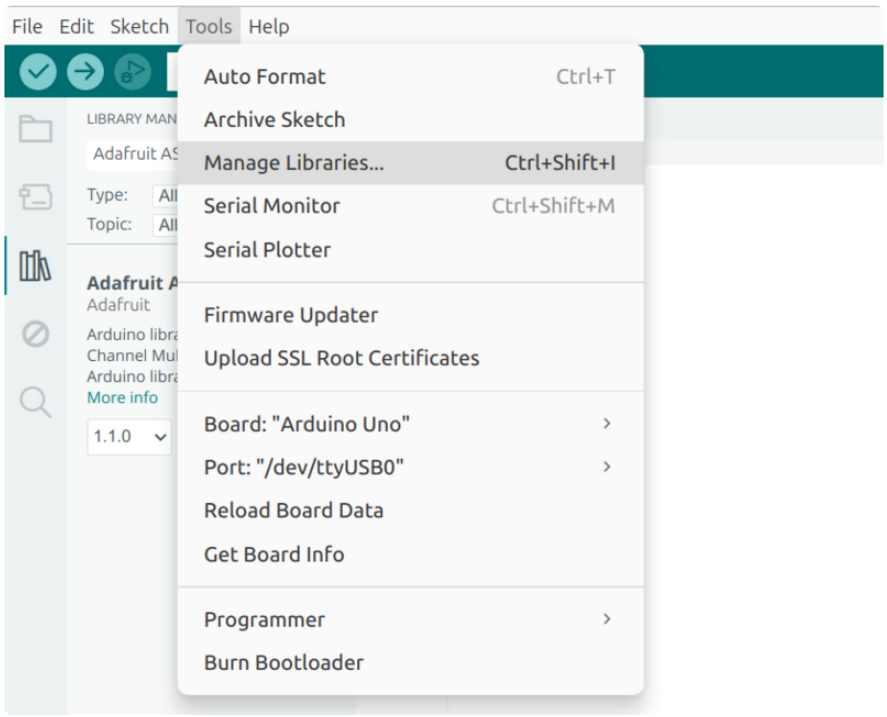
Here is an Adafruit Metro wired up using a solderless breadboard:



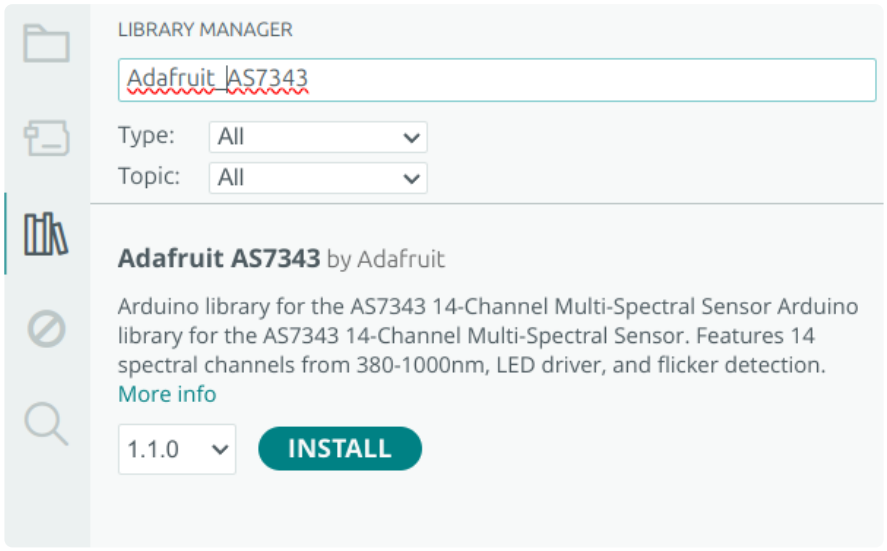
Board 5V to **breakout VIN (red wire)**
Board GND to **breakout GND (black wire)**
Board SCL to **breakout SCL (yellow wire)**
Board SDA to **breakout SDA (blue wire)**

Library Installation

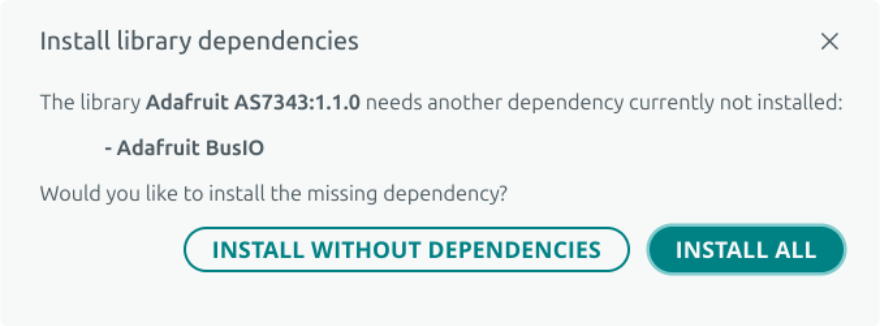
You can install the **Adafruit_AS7343** library for Arduino using the Library Manager in the Arduino IDE.



Click the **Manage Libraries ...** menu item, search for **Adafruit_AS7343**, and select the **Adafruit_AS7343** library:



If asked about dependencies, click "Install all".



If the "Dependencies" window does not come up, then you already have the dependencies installed.

e dependencies are already installed, you must make sure you update
n through the Arduino Library Manager before loading the example!

Example Code

```
/*!
 * @file basic_readings.ino
 *
 * Basic example for the Adafruit AS7343 14-Channel Multi-Spectral Sensor
 *
 * Reads all 18 channels (14 spectral + VIS + FD in 3 cycles) and prints
 * the values to Serial.
 */

#include <Adafruit_AS7343.h>

Adafruit_AS7343 as7343;

void setup() {
  Serial.begin(115200);
  while (!Serial) {
    delay(10); // Wait for Serial on native USB boards
  }

  Serial.println("Adafruit AS7343 Basic Readings");

  if (!as7343.begin()) {
    Serial.println("Could not find AS7343 sensor!");
    while (1) {
      delay(10);
    }
  }

  Serial.println("AS7343 found!");

  // Configure sensor
  as7343.setGain(AS7343_GAIN_64X);
  as7343.setATIME(29);  // Integration cycles
```

```
  as7343.setASTEP(599); // Step size

  Serial.print("Integration time: ");
  Serial.print(as7343.getIntegrationTime());
  Serial.println(" ms");
}

void loop() {
  uint16_t readings[18];

  // Read all channels (starts measurement, waits, reads internally)
  if (!as7343.readAllChannels(readings)) {
    Serial.println("Read failed!");
    delay(500);
    return;
  }

  // Print spectral channels (wavelength order)
  Serial.println("\n--- Spectral Readings ---");

  Serial.print("F1  (405nm violet):     ");
  Serial.println(readings[AS7343_CHANNEL_F1]);

  Serial.print("F2  (425nm violet-blue):");
  Serial.println(readings[AS7343_CHANNEL_F2]);

  Serial.print("FZ  (450nm blue):       ");
  Serial.println(readings[AS7343_CHANNEL_FZ]);

  Serial.print("F3  (475nm blue-cyan):  ");
  Serial.println(readings[AS7343_CHANNEL_F3]);

  Serial.print("F4  (515nm green):      ");
  Serial.println(readings[AS7343_CHANNEL_F4]);

  Serial.print("F5  (550nm green-yel):  ");
  Serial.println(readings[AS7343_CHANNEL_F5]);

  Serial.print("FY  (555nm yellow-grn): ");
  Serial.println(readings[AS7343_CHANNEL_FY]);

  Serial.print("FXL (600nm orange):     ");
  Serial.println(readings[AS7343_CHANNEL_FXL]);

  Serial.print("F6  (640nm red):        ");
  Serial.println(readings[AS7343_CHANNEL_F6]);

  Serial.print("F7  (690nm deep red):   ");
  Serial.println(readings[AS7343_CHANNEL_F7]);

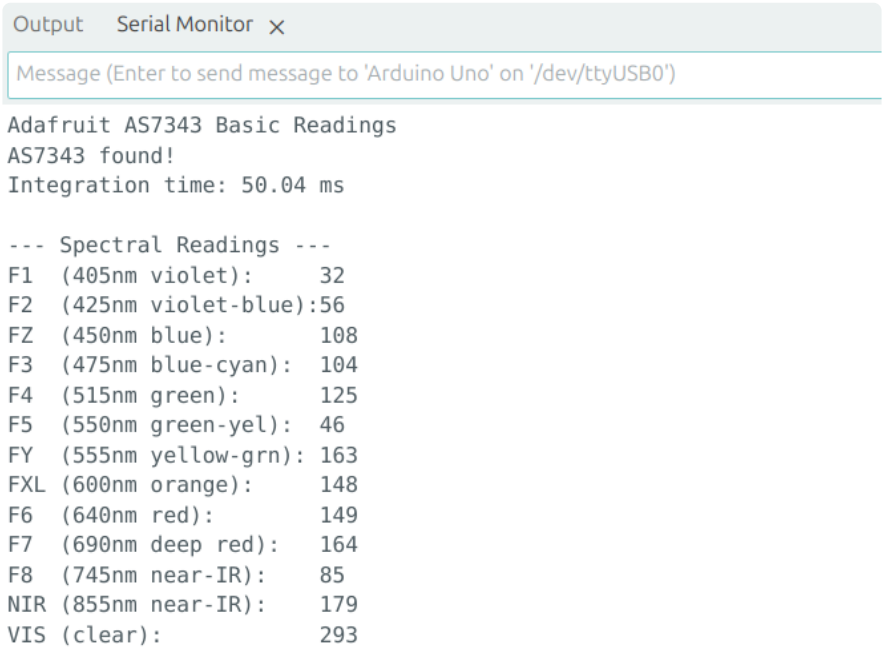
  Serial.print("F8  (745nm near-IR):    ");
  Serial.println(readings[AS7343_CHANNEL_F8]);

  Serial.print("NIR (855nm near-IR):    ");
  Serial.println(readings[AS7343_CHANNEL_NIR]);

  // Print clear/VIS channels (one from each cycle)
  Serial.print("VIS (clear):            ");
  Serial.println(readings[AS7343_CHANNEL_VIS_TL_0]);

  delay(500);
}
```

Upload the sketch to your board and open up the Serial Monitor (**Tools -> Serial Monitor**) at 115200 baud. You'll see the AS7343 recognized over I2C. Then, the spectral reading values of different channels will be printed out to the Serial Monitor.



Arduino Docs

Arduino Docs (https://adafru.it/1aAX)

Downloads

Files

- AS7343 Datasheet (https://adafru.it/1aBa)
- EagleCAD PCB files on GitHub (https://adafru.it/1aBb)
- Fritzing object in the Adafruit Fritzing Library (https://adafru.it/1aBc)

Schematic and Fab Print

